PHASE 8 – PROJECT DEMONSTRATION

AI FAQ Assistant API – AI-Powered FAQ and Customer Support Automation

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1. Introduction

Phase 8 is the **Project Demonstration** phase of the AI FAQ Assistant API project. This phase focuses on preparing and presenting a **demo video** that shows the working features of the developed application.

The demo video provides visual evidence of the completed project and demonstrates the major API functions using the development environment and Postman.

The video explains the project briefly and then demonstrates the important features in a clear sequence.

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2. Objective of the Demo Video

The main objectives of the demo video are:

1. To introduce the AI FAQ Assistant API project.
2. To explain the purpose of the application.
3. To show the project development environment.
4. To demonstrate user authentication.
5. To demonstrate FAQ management.
6. To demonstrate FAQ search.
7. To demonstrate AI answer generation.
8. To demonstrate AI FAQ generation.
9. To show the working API responses.
10. To provide final evidence of the completed project.

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3. Demo Video Preparation

Before recording the video, the project environment is prepared to ensure that the demonstration can be completed without unnecessary interruptions.

The following components are prepared:

- Node.js project.
- MongoDB database.
- Project source code.
- Required environment variables.
- Gemini API configuration.
- Postman application.
- AI FAQ Assistant API Postman collection.
- Test user account.
- Sample FAQ data.

The backend server is started before beginning the main API demonstrations.

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4. Demo Video Structure

The demo video is organized into the following sequence:

Sequence	Demonstration
1	Project title and introduction
2	Problem and proposed solution
3	Technologies used
4	Project structure
5	Backend server startup
6	User registration
7	User login and JWT authentication
8	FAQ management
9	FAQ search
10	AI answer generation
11	AI FAQ generation

Sequence	Demonstration
12	Error/security demonstration
13	Final project summary

This sequence allows the viewer to understand the project before seeing its major working features.

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5. Project Introduction in the Video

The video begins with a short introduction to the project.

The following information is presented:

Project Title:
AI FAQ Assistant API – AI-Powered FAQ and Customer Support Automation

The introduction explains that the project is a RESTful backend API designed to manage FAQs and provide AI-powered customer support functionality.

The technologies used in the project are briefly mentioned:

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT
- bcrypt
- Google Gemini API
- Postman

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6. Project Setup Demonstration

The demo video shows the project environment before demonstrating the API functionality.

The project folder structure can be briefly displayed to show the organization of the source code.

The video then shows the Node.js server being started.

The server runs using the configured local address:

http://localhost:5000

This confirms that the backend application is running and ready for demonstration.

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7. Authentication Demonstration

The first major functional demonstration is user authentication.

Registration

The video demonstrates the registration API:

POST /api/auth/register

A new user is registered through Postman and the API response is shown.

Login

The login API is then demonstrated:

POST /api/auth/login

After successful login, the server returns a JWT token.

The video shows that the token can be used to access protected API operations.

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8. FAQ Management Demonstration

The video next demonstrates the FAQ management functionality.

The following operations are shown:

Create FAQ

POST /api/faqs

A new FAQ is created using an authenticated request.

Retrieve FAQs

GET /api/faqs

The available FAQ records are displayed.

Retrieve Individual FAQ

GET /api/faqs/:id

A specific FAQ is retrieved using its ID.

Update FAQ

PUT /api/faqs/:id

The FAQ information is modified.

Delete FAQ

DELETE /api/faqs/:id

The selected FAQ is removed.

The video does not need to demonstrate every CRUD operation in excessive detail; the important requests and responses are shown clearly.

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9. FAQ Search Demonstration

The video demonstrates the keyword-based FAQ search functionality.

The API endpoint used is:

GET /api/faqs/search?q=query_string

For example:

GET /api/faqs/search?q=MongoDB

The response is displayed in Postman to show the FAQs matching the supplied keyword.

This demonstrates how users can retrieve relevant FAQ information without manually checking every FAQ record.

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10. AI Answer Generation Demonstration

The AI Answer Generator is one of the main features highlighted in the demo video.

The following API is demonstrated:

POST /api/ai/answer

A sample question is entered into the request.

For example:

```
{
  "question": "What is MongoDB?"
}
```

The request is sent to the backend, which processes it through the Gemini service.

The generated AI response is displayed in Postman.

The video briefly explains the processing flow:

Question → API → AI Controller → Gemini Service → AI Answer

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11. AI FAQ Generation Demonstration

The second AI feature demonstrated is AI-powered FAQ generation.

The API endpoint is:

POST /api/ai/generate-faq

A topic is provided as input.

Example:

```
{
  "topic": "MongoDB"
}
```

The backend sends the request to the configured Gemini service and returns the generated FAQ information.

The response is shown clearly in the demo video.

This demonstrates how AI can assist in creating FAQ content based on a supplied topic.

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12. Security and Error Demonstration

A short section of the video demonstrates the application's security and error-handling capabilities.

One or two simple examples can be shown, such as:

- Accessing a protected endpoint without a JWT token.
- Attempting to register an already existing email.
- Providing an invalid FAQ ID.

The resulting API error response is displayed.

This demonstrates that the application does not only handle successful requests but also provides responses for invalid or unauthorized requests.

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13. Demo Video Recording and Editing

The project demonstration is recorded as a screen-capture video.

During recording:

- The required applications are opened beforehand.
- Unnecessary windows are closed.
- The project server is started before the main demonstration.
- Postman requests are arranged in a logical order.
- Important API responses are kept visible long enough to be understood.
- The demonstration is kept focused on the project's actual features.

After recording, unnecessary pauses or repeated sections can be removed during basic video editing.

The final video should maintain a clear flow from **project introduction → setup → functionality → AI features → conclusion**.

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14. Final Demo Video Sequence

The final video follows this sequence:

Project Introduction

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Problem and Proposed Solution

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Technology Stack

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Project Structure

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Server Startup

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User Registration

↓

User Login + JWT

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FAQ Management

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FAQ Search

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AI Answer Generation

↓

AI FAQ Generation

↓

Security / Error Handling

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Project Summary

This sequence ensures that the viewer can understand both the purpose and working functionality of the project.

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15. Final Project Demonstration Outcome

The completed demo video provides visual evidence of the working **AI FAQ Assistant API**.

The demonstration covers the major implemented functions:

Feature	**Demonstrated**
Backend Server	✓
User Registration	✓
User Login	✓
JWT Authentication	✓
FAQ Creation	✓
FAQ Retrieval	✓
FAQ Update/Delete	✓

Feature	Demonstrated
FAQ Search	✓
AI Answer Generation	✓
AI FAQ Generation	✓
Error Handling	✓

The demo video therefore serves as a concise presentation of the completed project.

DEMO VIDEO DRIVE LINK:
https://drive.google.com/file/d/1BT9QVYbzRTQOvwVcTje9HqMx344uxlbo/view?usp=sharing

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16. Project Demonstration Conclusion

The **Project Demonstration phase** completes the eight-phase development process of the **AI FAQ Assistant API**.

The demo video presents the project in a structured sequence, beginning with the project introduction and ending with the demonstration of its major working features.

The video provides practical evidence of the implemented authentication, FAQ management, search, AI integration, and security functionality.

The final demonstration represents the completed project and provides a clear visual presentation of the work carried out during the previous development phases.